SOFTWARE DESIGN DESCRIPTION

Daily Deals & Discounts

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Revision History

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30-10-2026	1.0	Zunaira Kanwal	Initial SDD Document Release

Document Approval

The Software Design Description has been accepted and approved by the following supervisor:

Signature	Printed Name	Title	Date
	Dr. Muhammad Dost Khan	Supervisor, IT Department	30-10-2026

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1. INTRODUCTION

1.1 Purpose

This Software Design Document (SDD) describes the architecture and detailed design of the Daily Deals & Discounts system. It presents internal structure, components, data design, and interaction mechanisms of the web application for developers, project supervisors, and testers.

1.2 Scope

Daily Deals & Discounts helps customers discover nearby promotional offers while enabling vendors to advertise deals. Administrators moderate user accounts and approve deal submissions.

1.3 Reference Material

- PakWheels Classifieds (2026) - https://www.pakwheels.com
- OLX Pakistan (2026) - https://www.olx.com.pk
- React Documentation - https://react.dev
- Node.js Documentation - https://nodejs.org
- IEEE Software Design Description Standards

1.4 Definitions and Acronyms

Term / Acronym	Description
Deal / Vendor	Promotional offer posted by business users
Customer / Admin	System users browsing deals and managers moderating platform
SRS / SDD	Software Requirements Specification / Software Design Document
MVC / API	Model View Controller Architecture / Application Programming Interface

2. SYSTEM OVERVIEW

Daily Deals & Discounts is a web-based marketplace operating with three core modules: Customer Module, Vendor Module, and Admin Module. The architecture is built upon the Model-View-Controller (MVC) pattern utilizing relational data structures.

3. SYSTEM ARCHITECTURE

3.1 Architectural Design

The system follows the traditional MVC structure:

- **Models:** User, Vendor, Admin, Deal, Category, Review, Report, Notification.
- **Controllers:** User Controller, Vendor Controller, Admin Controller, Deal Controller.
- **Views:** Home, Login/Signup, Deal Listing, Vendor Dashboard, Admin Dashboard.

3.2 Subsystem Decomposition

- **Customer Subsystem:** Registration, deal searching, saving offers, submitting reviews and reports.
- **Vendor Subsystem:** Deal creation, editing, deletion, and analytics dashboard.
- **Admin Subsystem:** Account approvals, content moderation, category and user management.

4. DATA DESIGN

4.1 Data Dictionary

Entity	Attributes	Methods
User	UserID, FullName, Email, PasswordHash, PhoneNumber, Location, UserRole	SignUp(), Login(), UpdateProfile()
Vendor	VendorID, UserID, BusinessName, ApprovalStatus	PostDeal(), EditDeal(), DeleteDeal(), ViewAnalytics()
Deal	DealID, Title, Description, OriginalPrice, DiscountedPrice, ExpiryDate, Status	CreateDeal(), UpdateDeal(), DisplayDeal()
Admin	AdminID, Username, PasswordHash	ApproveDeal(), RejectDeal(), ManageUsers()

5. COMPONENT DESIGN

Authentication & Deal Approval Logic

```
      // Authentication Component Pseudo Code START Input Email and Password Validate Inputs
IF Credentials Correct THEN Create User Session ELSE Display Error Message END // Deal
Approval Component Pseudo Code START Admin selects Pending Deal Review deal details & images
IF Valid THEN Update status = Approved ELSE Update status = Rejected Send notification to
Vendor END
```

6. HUMAN INTERFACE DESIGN

The interface is responsive, accessible across desktop and mobile browsers, featuring clear navigation for deal discovery and dashboard operations.

7. REQUIREMENTS MATRIX

Req ID	Description	Module
FR01	User Registration & Authentication	Authentication Module
FR02	Profile Management	User Module
FR03	Post, Edit, Delete Deal	Deal Management Module
FR04	Search & Geo-Filter Deals	Search Module
FR05	Save Deals & Submit Reviews	Customer Module
FR06	Approve/Reject Deals & Moderation	Admin Module
FR07	Vendor Analytics Dashboard	Dashboard Module

8. APPENDICES

Appendix A — Technologies Used

React.js, Node.js, Express.js, MongoDB / Relational Database, REST APIs, Google Maps API, MVC Architecture.

Appendix B — Project Modules

- **Customer Module:** Browsing, filtering, saving, and reviewing deals.
- **Vendor Module:** Posting deals, managing inventory, viewing response analytics.
- **Admin Module:** Approval workflow, moderation, user management.

End of Document — Department of Information Technology, The Islamia University of Bahawalpur